

Module 4: Data quality, measurement error, and missing data

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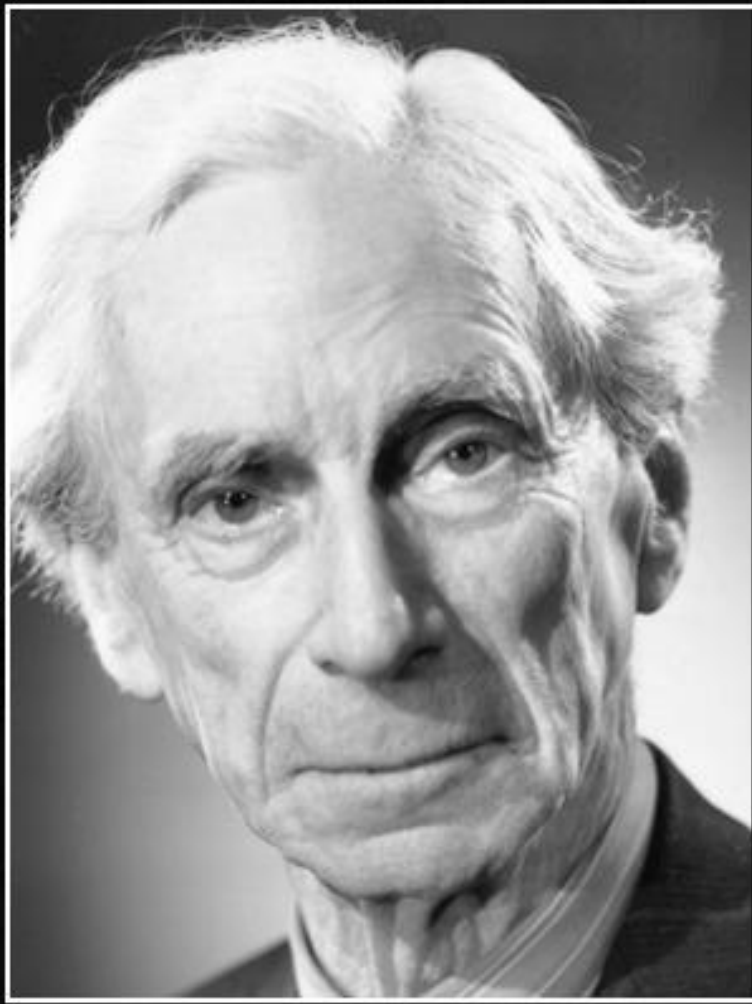
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Outline of this module

- **Introduction**
- Measurement error
- Mis-reporting of dietary intake
 - Under-reporting
- Missing data
 - Non-response
- Conclusion



Dietary intake can only rarely be directly observed



The observer, when he seems to himself to be observing a **DIET** is really, if physics is to be believed, observing the effects of the **DIET** upon himself.

— *Bertrand Russell* —

When dietary intake is observed, it may change

By the end of this module, you will be able to:

- **Define** key terms
 - Measurement error, non-response, missing data, sampling error, non-sampling error
- List **three threats** to data quality for dietary recall data
- Classify **three types** of bias in dietary recall data
- Identify **three ways** to improve data quality for dietary intake surveys
- **Interpret results** from dietary recall data in the context of missing data or bias
- **Advocate** for the necessity of dietary recall data despite limitations

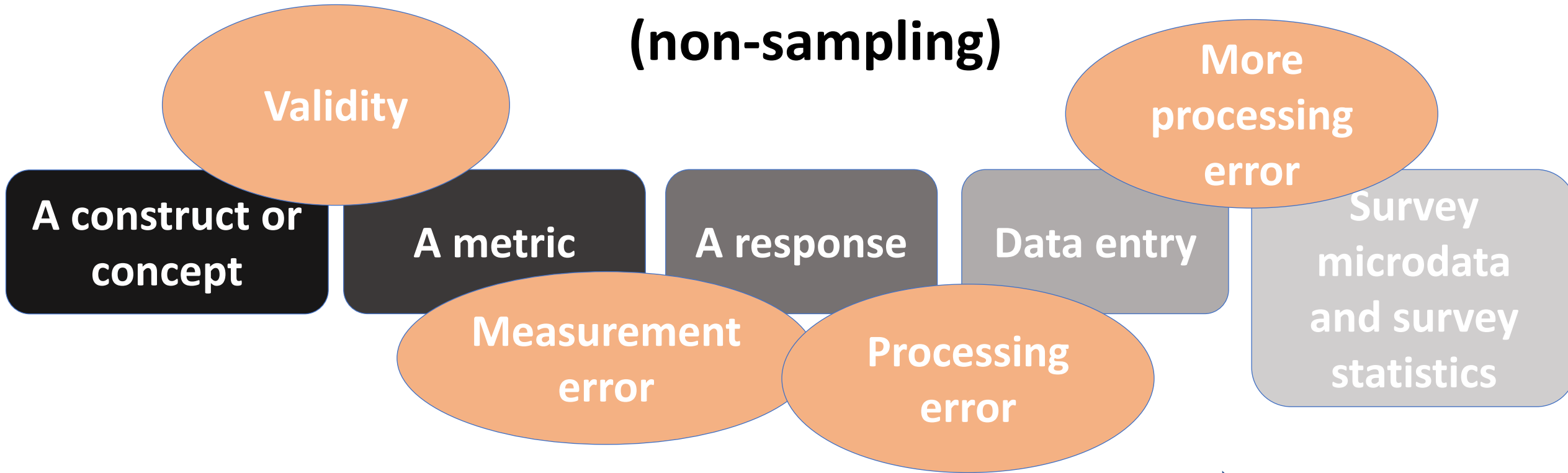


Defining data quality

- **There are many possible definitions**
 - The definition for you as a researcher depends on *your* specific project, research question, and data used.
 - A good definition would include issues related to:
 - Completeness
 - Systematic errors (bias)
 - Variance (noise)
 - How the data can be used after they are collected (utility)

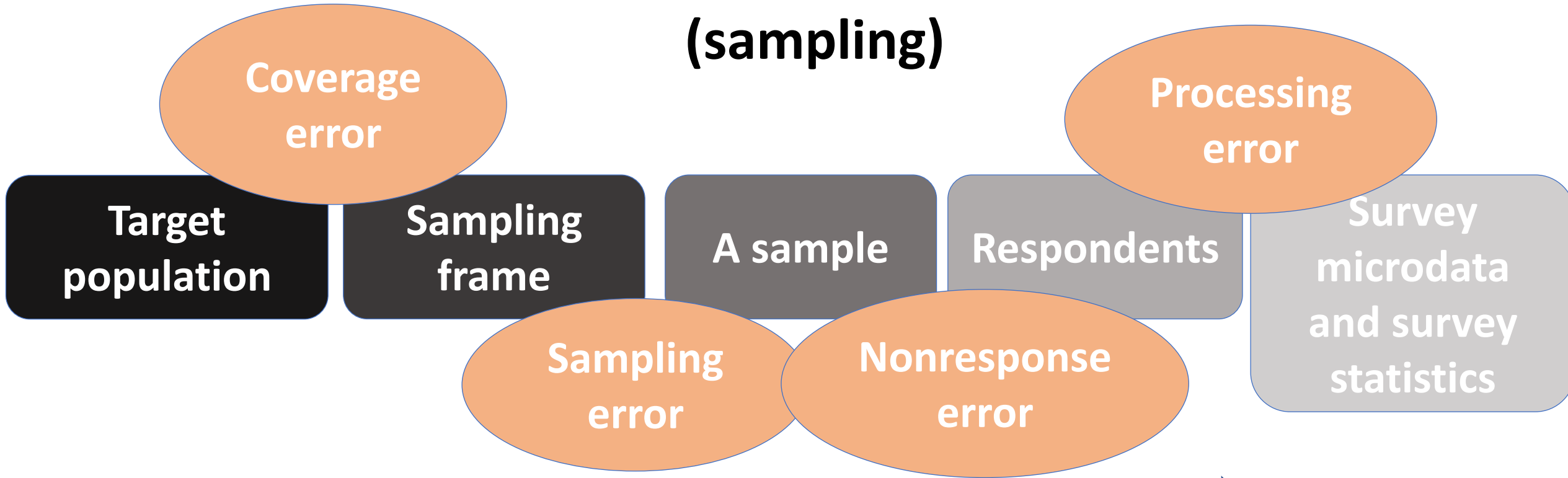


Errors in measurement (non-sampling)



There are many spots where data quality can be reduced on the path from 'concept' to final data

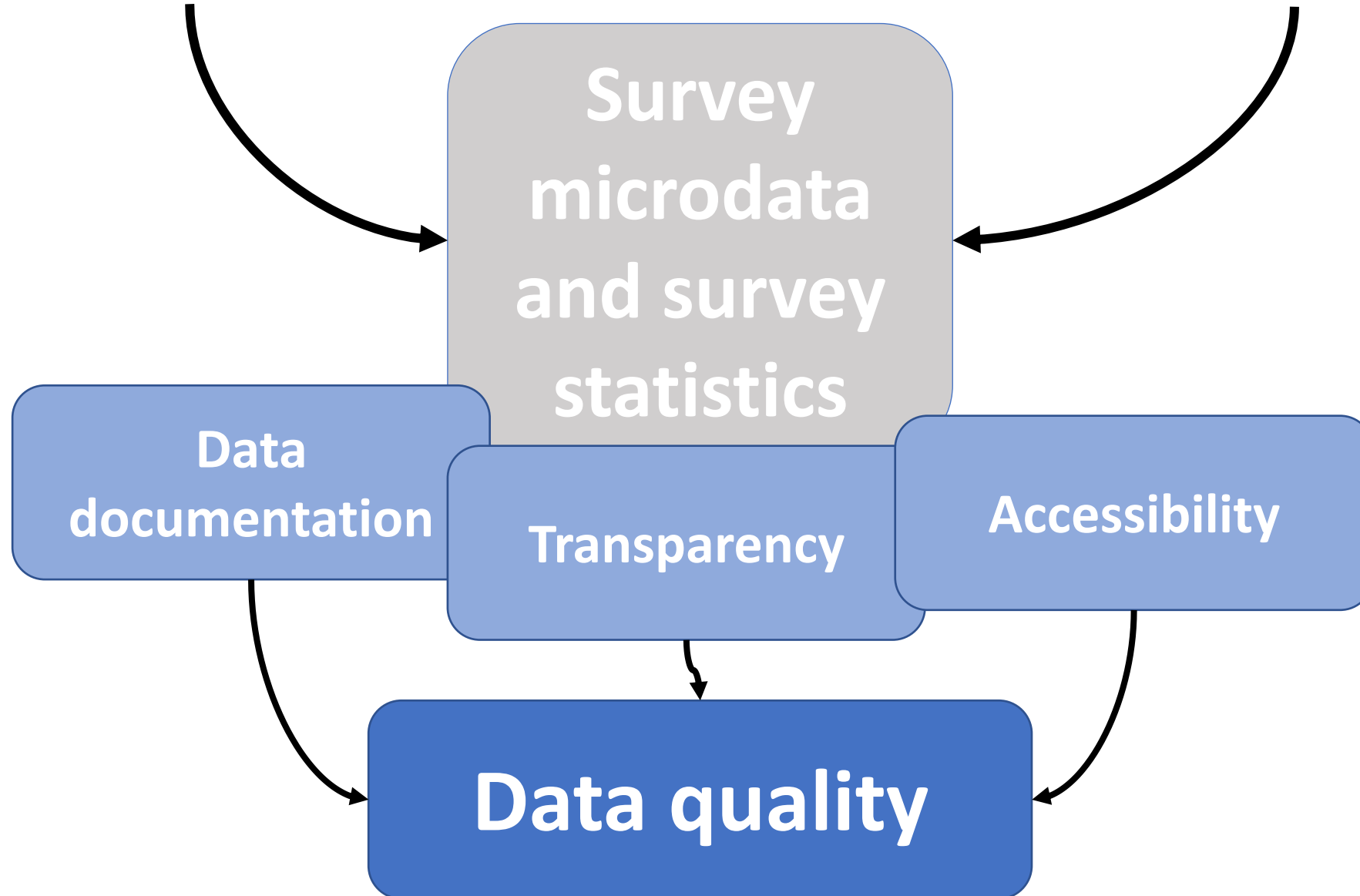
Errors in representativeness (sampling)



There are many spots where data quality can be reduced on the path from 'concept' to final data

**Errors in representativeness
(sampling)**

**Errors in measurement
(non-sampling errors)**



Types of survey data quality issues

- **Sampling errors**

- Seemingly random; this happens even if no mistakes were made
- Can be reduced if sample size is increased
- Biases an estimate of an association *towards* the null hypothesis
- Increase likelihood of a Type II error; wider confidence intervals
- Random measurement error can reduce the impact of a study for funders or policymakers

- **Non-sampling errors**

- Can be systematic or random
- Harder to estimate/know about than sampling error
- Hard to understand the underlying cause of
- Causes deflation towards the null, or inflation away from the null
- Most data quality research focuses on sampling errors



Defining data quality

Q: *What does “observable” mean?*

- **A high-quality dataset has:**
 - a high degree of completeness
 - an acceptable amount of random noise
 - no differences in precision or accuracy across any observable characteristics
 - the potential to answer your research question



Types of data quality issues

- Dataset(s) might have missing values (lack of *completeness*)
 - Data might be missing at random, or not
 - Reduces sample size and increases standard errors
 - Can be caused by **non-response**
- Dataset(s) might have incorrect values (*bias* and *noise*)
 - Some incorrect values can be spotted, others not
 - For health data, biological plausibility is helpful here
 - Can be caused by **underreporting**, overreporting, **measurement error**, or memory lapses



Data quality characteristics addressed by Scotland's National Statistics Office - 1

- Relevance
 - Does the research output meet users' needs?
- Timeliness and punctuality
 - Is the lapse of time between data collection and publication sufficiently short?
- Comparability
 - Can the data be compared over time and across geographic levels?
- Coherence
 - Are data about the same topic (like food) in different ways sufficiently similar?



Data quality characteristics addressed by Scotland's National Statistics Office - 2

- Accuracy
 - Is the degree of closeness between an estimate and the true value sufficiently small?
- Output quality trade-offs
 - Are different dimensions of data quality balanced against each other?
- Assessment of user needs and perceptions
 - Is the process for finding out about data users and their views and needs sufficient?
- Accessibility and clarity
 - Can users access the data? Is the data documentation clear?



There are benefits and challenges to all kinds of diet assessment instruments

Benefits with a 24-hr recall

- Detailed info is possible
 - Brands
 - Amounts
 - Cooking method
- Types of foods/beverages can be assessed together
- Can be linked with nutrient composition database
- Timing of meals and snacks
- High utility for nutritional epi

Challenges with a 24-hr recall

- Selection bias in who does the recalls
- Need >1 recall to assess variation across days
- Relies on *specific* memory
- What degree of choice do respondents have for when they fill out the recall?
- People do not excel at estimating the amount of food they ate, even with visual aids



Imagine you are in Intake24, doing your best to complete a recall.

- What might affect your ability to complete the recall accurately?



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What is *measurement error*?

- **Measurement error** is the difference between reported or estimated dietary intake over a time period and the **true** usual dietary intake.
- For diets, long-term average intake is more useful than short-term
 - For most nutritional epi purposes
- Data on dietary assessment can be prone to error, regardless of how careful the respondents and researchers are
- Measurement error affects scientific and observational research on all topics, not just diets.
 - But everyone **loves** to complain about how “inaccurate” dietary intake studies are.



Types of measurement error in diet assessment

- **Random error**

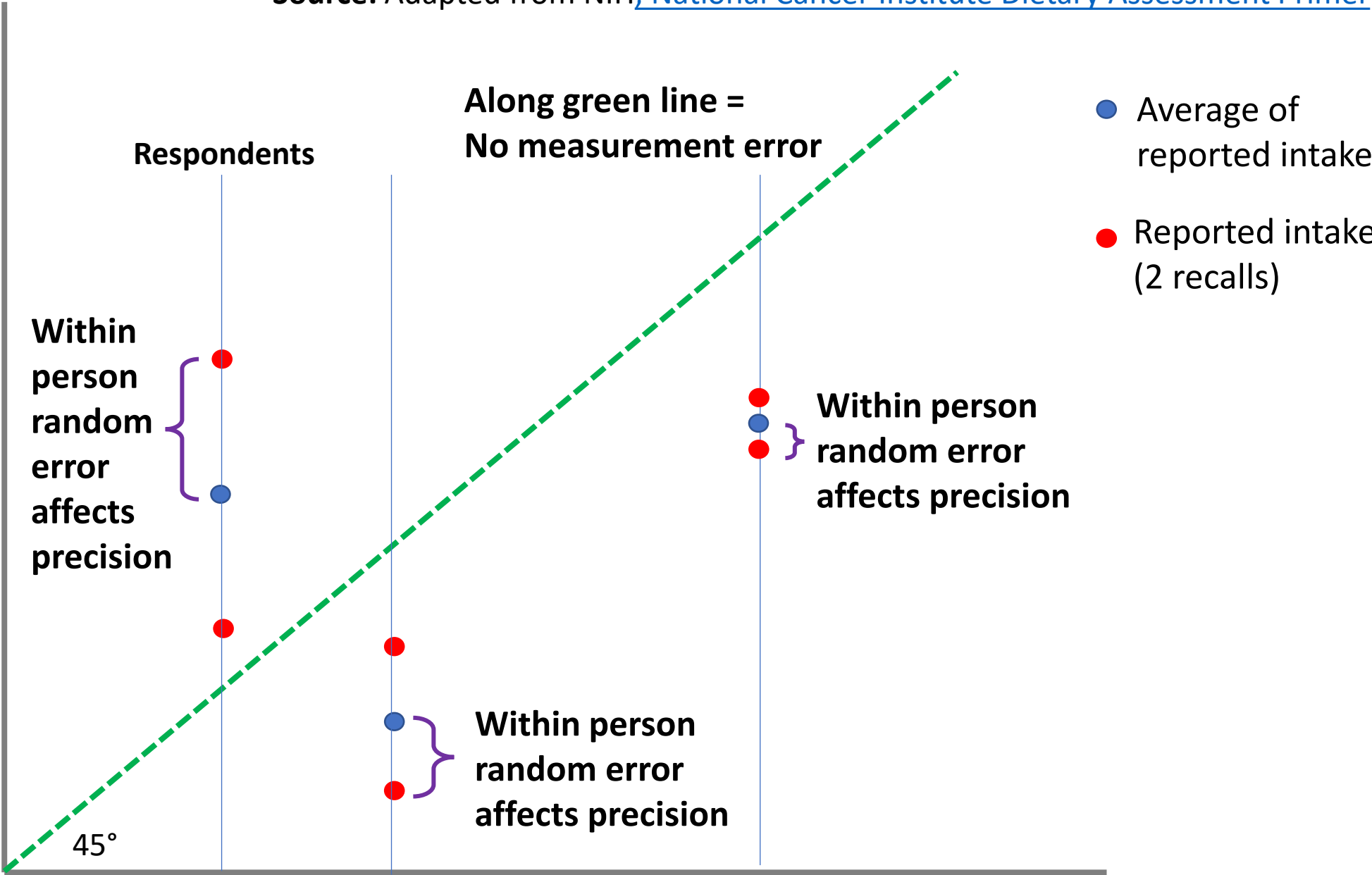
- Difference between estimates of dietary intake from one recall versus multiple administrations of a recall

- **Systematic error**

- Error can be related to true intake
- Error can be related to the characteristics of individual respondents
- Error can be related to the food item being asked about



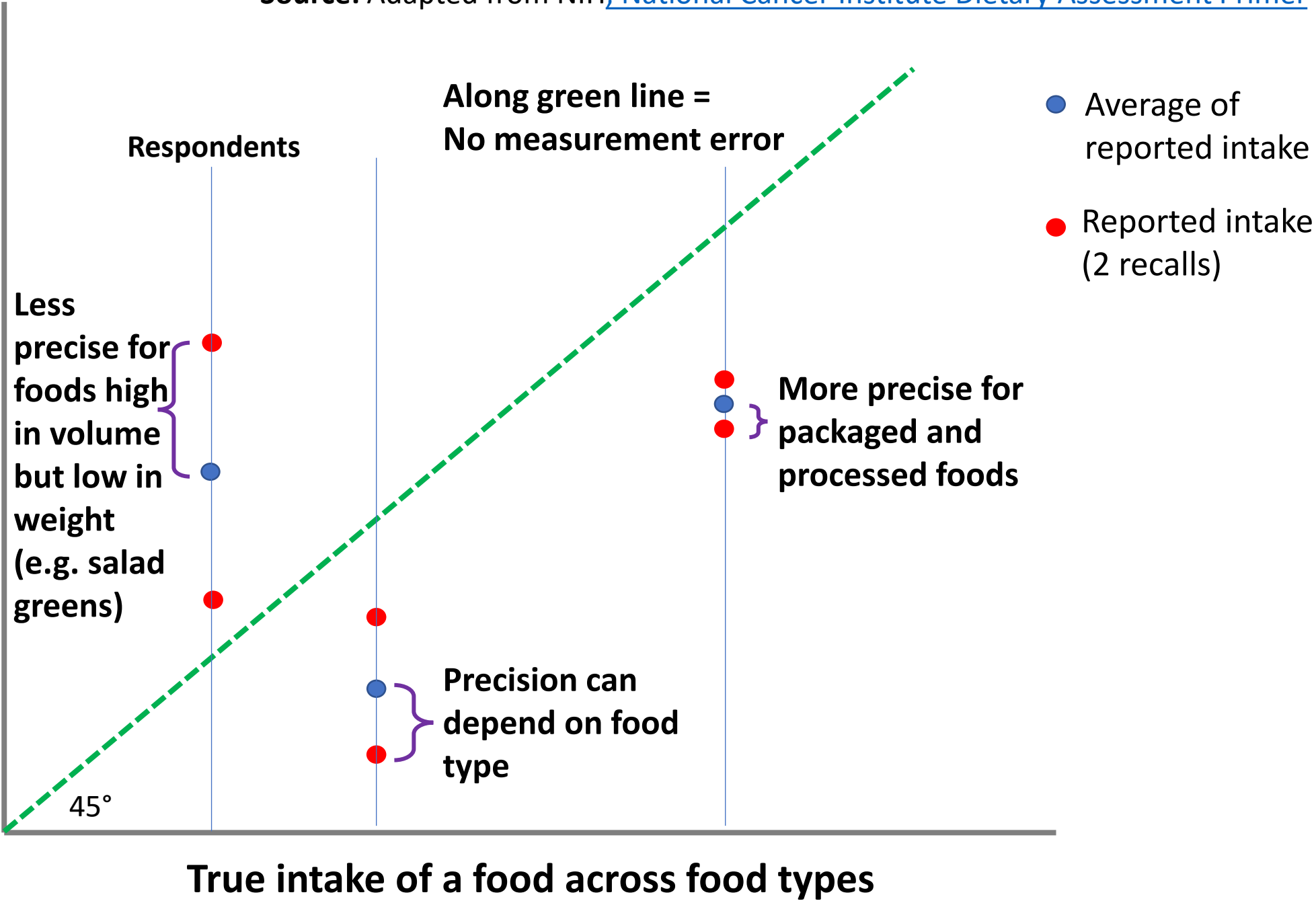
Reported intake
of a food



True intake =
reported intake

Reported intake
of a food

True intake =
reported intake

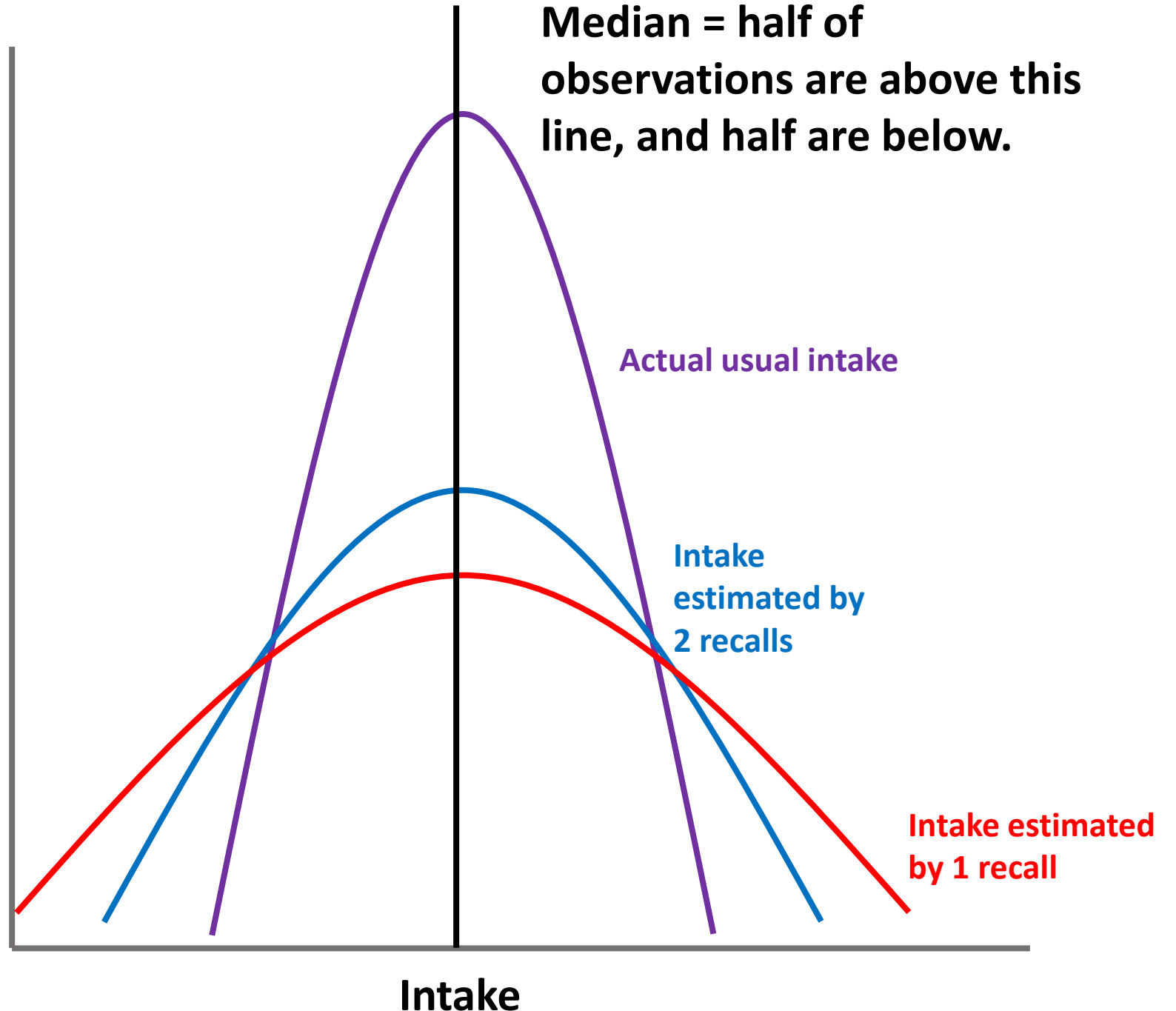


Density
= proportions of
the observations

Random measurement error
increases the width of estimated
intake distributions (increases the
confidence interval) but does not
change the median.

This matters for estimating the %
of the population above or below
a threshold, but not for estimating
mean usual intake.

**Median = half of
observations are above this
line, and half are below.**



Reported intake
of a food

Along green line =
No measurement error

● Averages of
reported intake

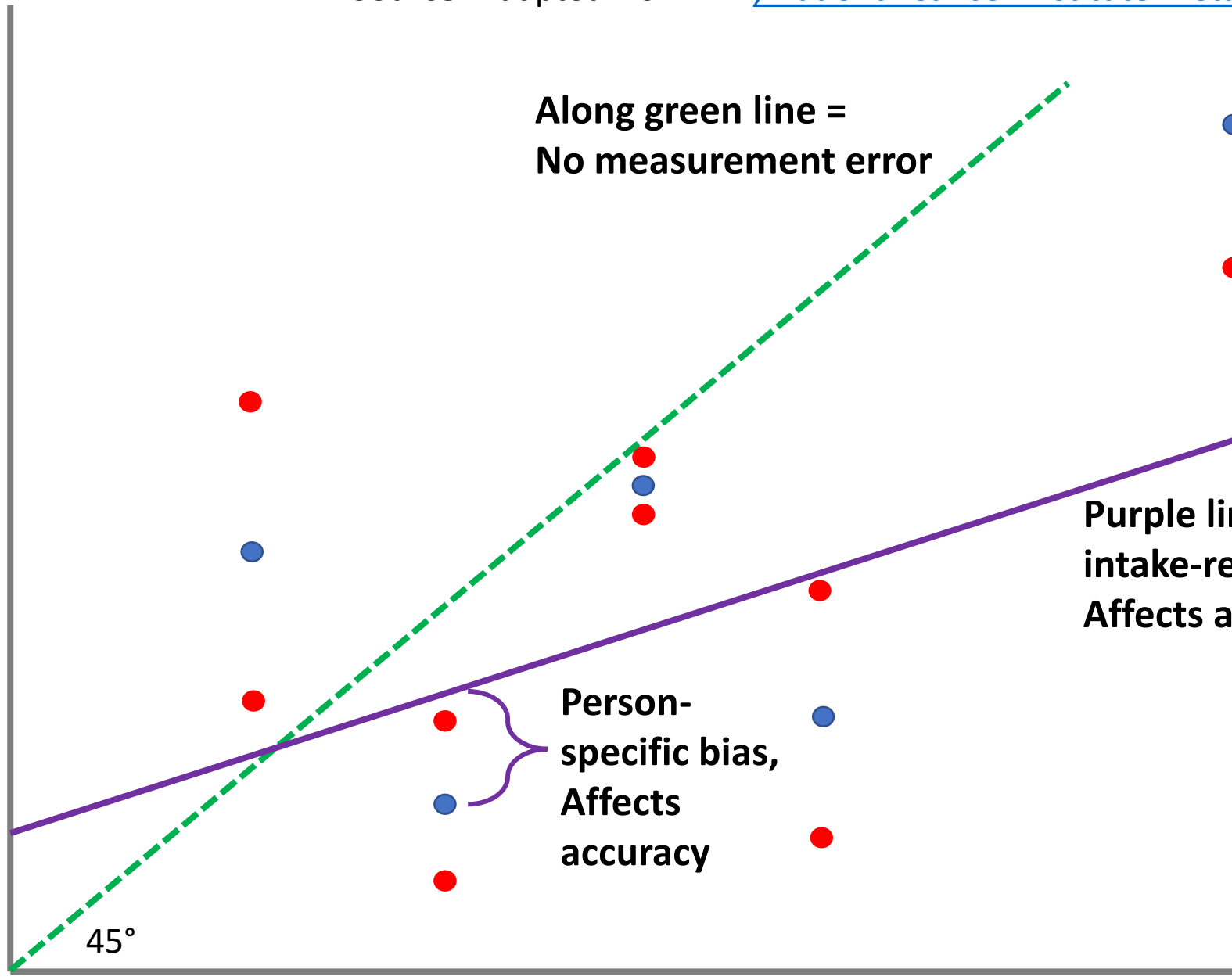
● Reported intake
(2 recalls)

Purple line =
intake-related bias
Affects accuracy

People with
higher true intake
tend to
underreport
intake

Person-
specific bias,
Affects
accuracy

True intake =
reported intake



45°

True intake of a food

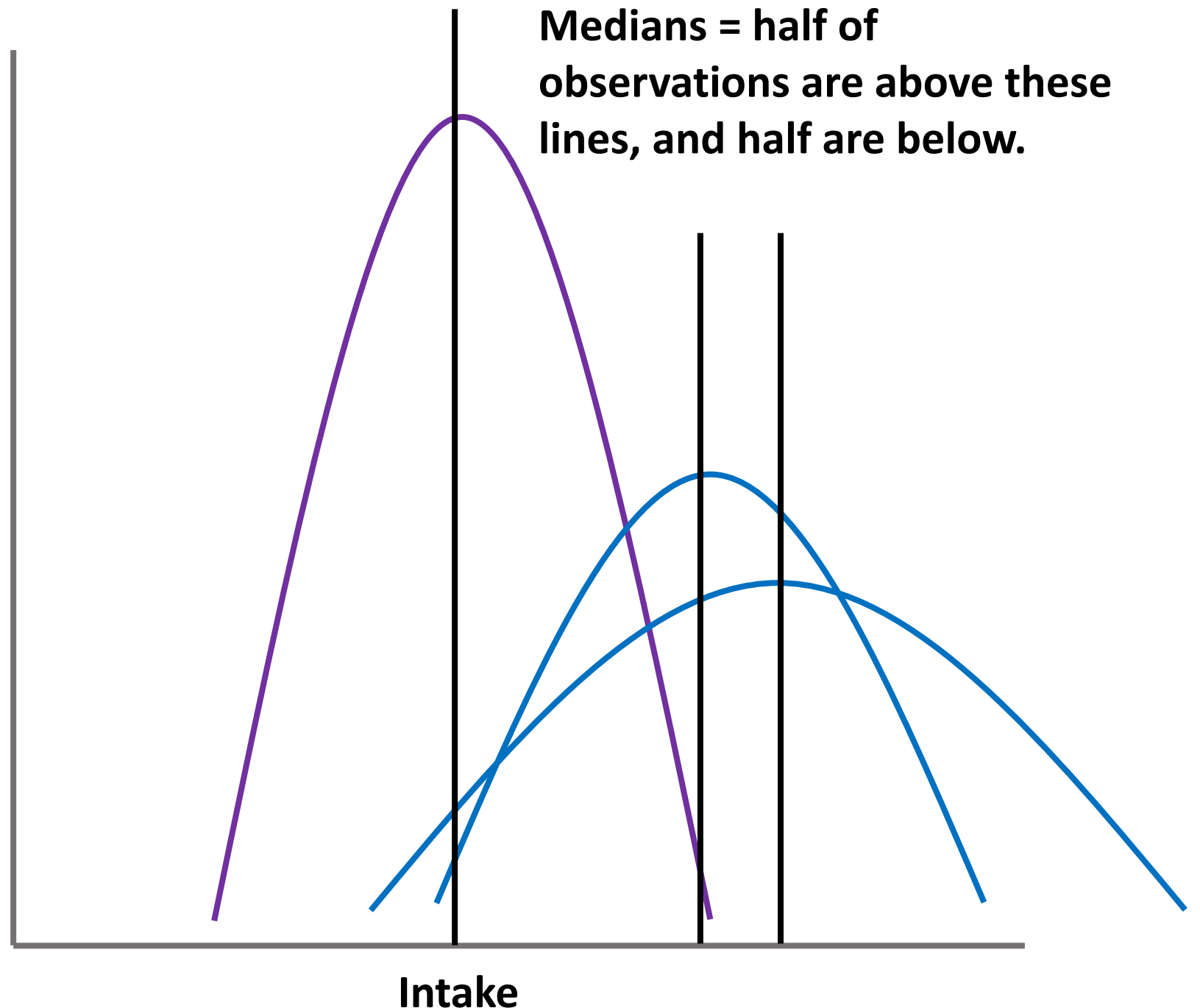
**Density
= proportions of
the observations**

Systematic measurement error can change the median estimated intake and the shape of the intake distribution.

Systematic bias can affect the median/mean and estimates of the % above or below a threshold.

24-hr recalls can have less bias than other methods of dietary assessment.

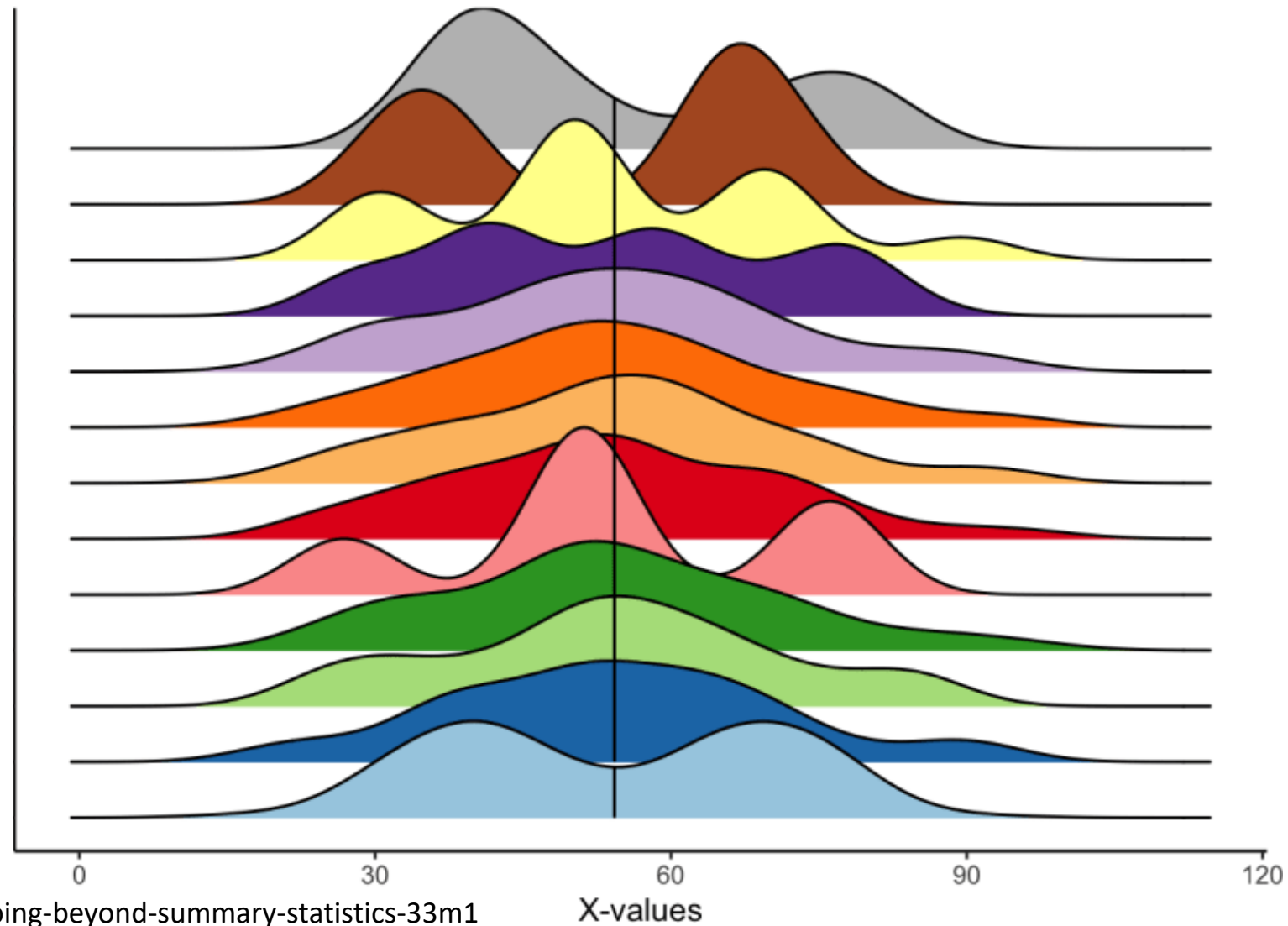
***Q:** What is the difference between estimating mean intake and % above or below a threshold? Is one metric more important than the other?*



Different distributions can have the same mean

Ridgeline Plot: More variation in the distribution (X-axis)

Identical mean values



Specific known systematic errors in dietary data

- The effect of **season**
 - Are some foods more likely to change more with season than others?
- The effect of **day-of-week**
 - Are some foods more likely to change with day-of-week than others?
- **Social desirability** bias
 - Healthy diets are (in part) a social construct – does this affect our recalls?
- **Memory lapses**
 - Is cognition related to dietary intake? Is cognition related to recall accuracy?
- **The observer effect**
 - Could participating in a recall change behavior?



Qs: Imagine you are a respondent to a study which uses Intake24, thinking about your own data.

- How does your diet change throughout the week?
- Do you eat differently on the weekends versus the weekdays?
- How does your diet change throughout the seasons?
- How might diet recall change with age, sex, region, geography?
- Is it possible to “control” for all factors which might affect person-specific bias?
- Might what you view as socially desirable affect how you report your own diet?
 - Does this depend on who is asking?



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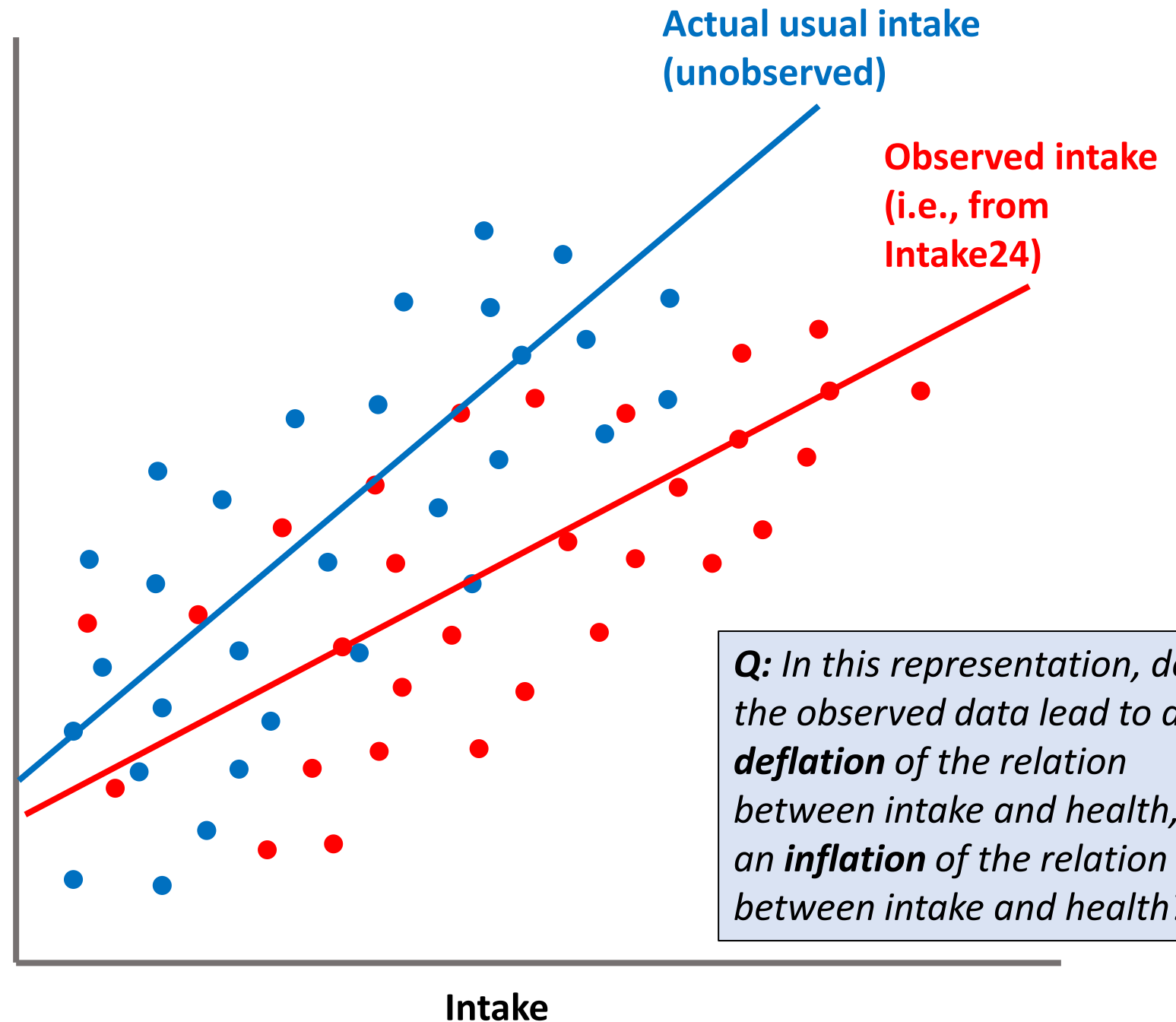
Misreporting is a demonstrated issue, especially for *energy* intake

- Intentional under- or over-reporting
- Unintentional underreporting
 - Associated with various respondent characteristics, including BMI
 - Food being eaten but genuinely forgotten
 - Beverages!
- Unintentional overreporting
 - Fruits and vegetables more likely to be overreported
 - Reporting of foods that were not actually consumed
- Changing food intake during the study period
 - Observer effect/Hawthorne effect



Health
outcome
of interest

When estimating the association between intake and a health outcome in the context of measurement error, the association may be inflated or deflated depending on sources of bias and random measurement error.



Consequences of measurement error in dietary assessment

- **Random error**

- Deflation of estimated relationships toward the null hypothesis
- Loss of statistical power
- Distributions are too wide

- **Systematic error**

- Exaggerated associations between health and diet
- Deflation of estimated relationships toward the null hypothesis
- Bias away from the null hypothesis
- Loss of statistical power



Solutions to types of measurement error

- **Random error**

- More repeated measures, averaging across days

- **Systematic error**

- Can only fix with a reference indicator, e.g., a biomarker.
 - Doubly-labeled water for measuring energy intake
 - Comparing with same-day weighed intakes

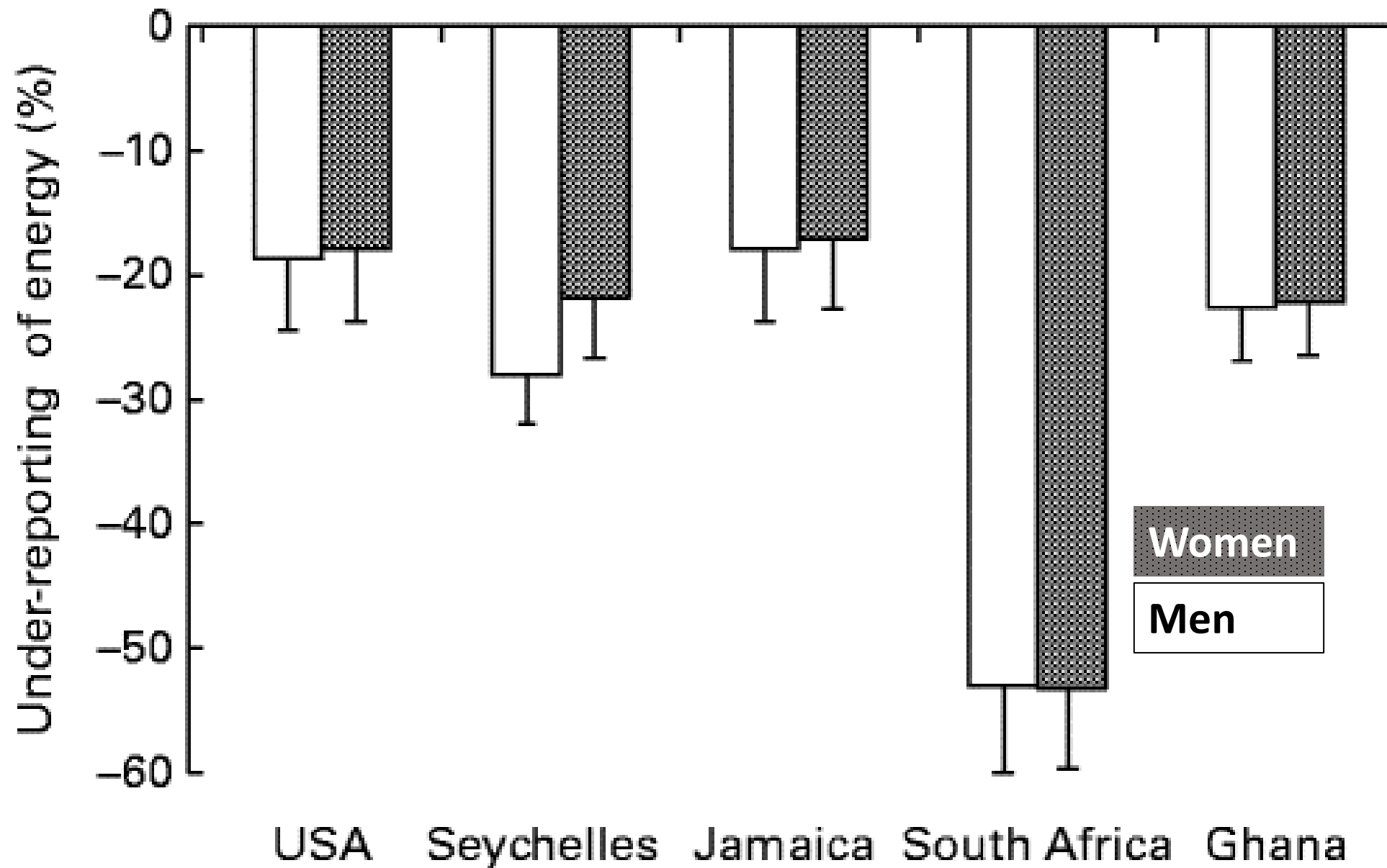


Under-reporting of foods can happen in patterns

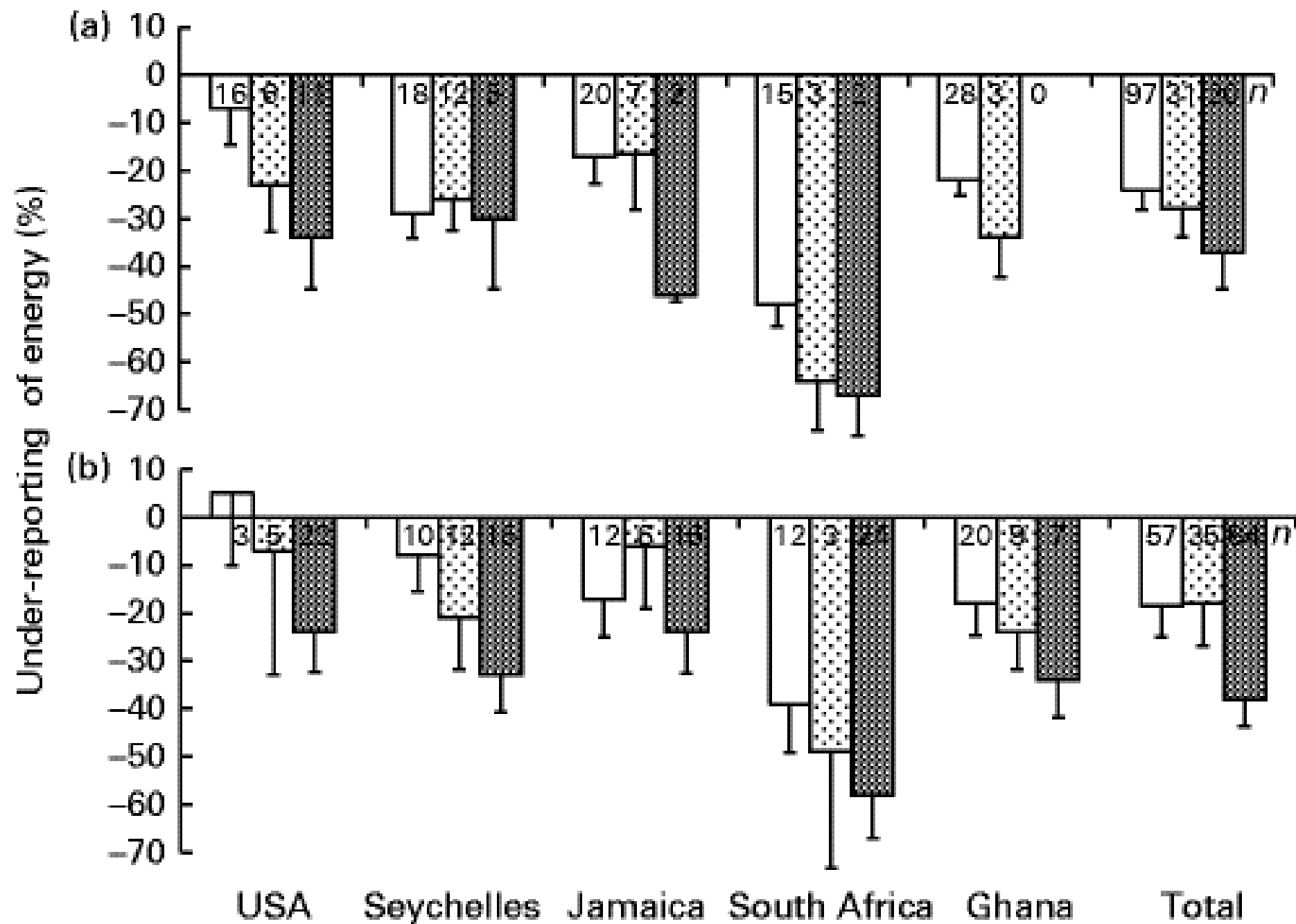
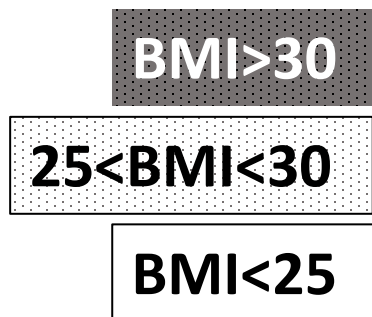
- Underreporting occurs **more frequently** for afternoon snacks and condiments at breakfast
- Underreporting and overreporting **patterns are different** within populations
- People with larger BMIs **may** underreport more
- Women **may** underreport more

Q: When analyzing dietary data, would you be more concerned with random underreporting across your sample, or underreporting that was larger in magnitude for certain sub-groups within your sample?

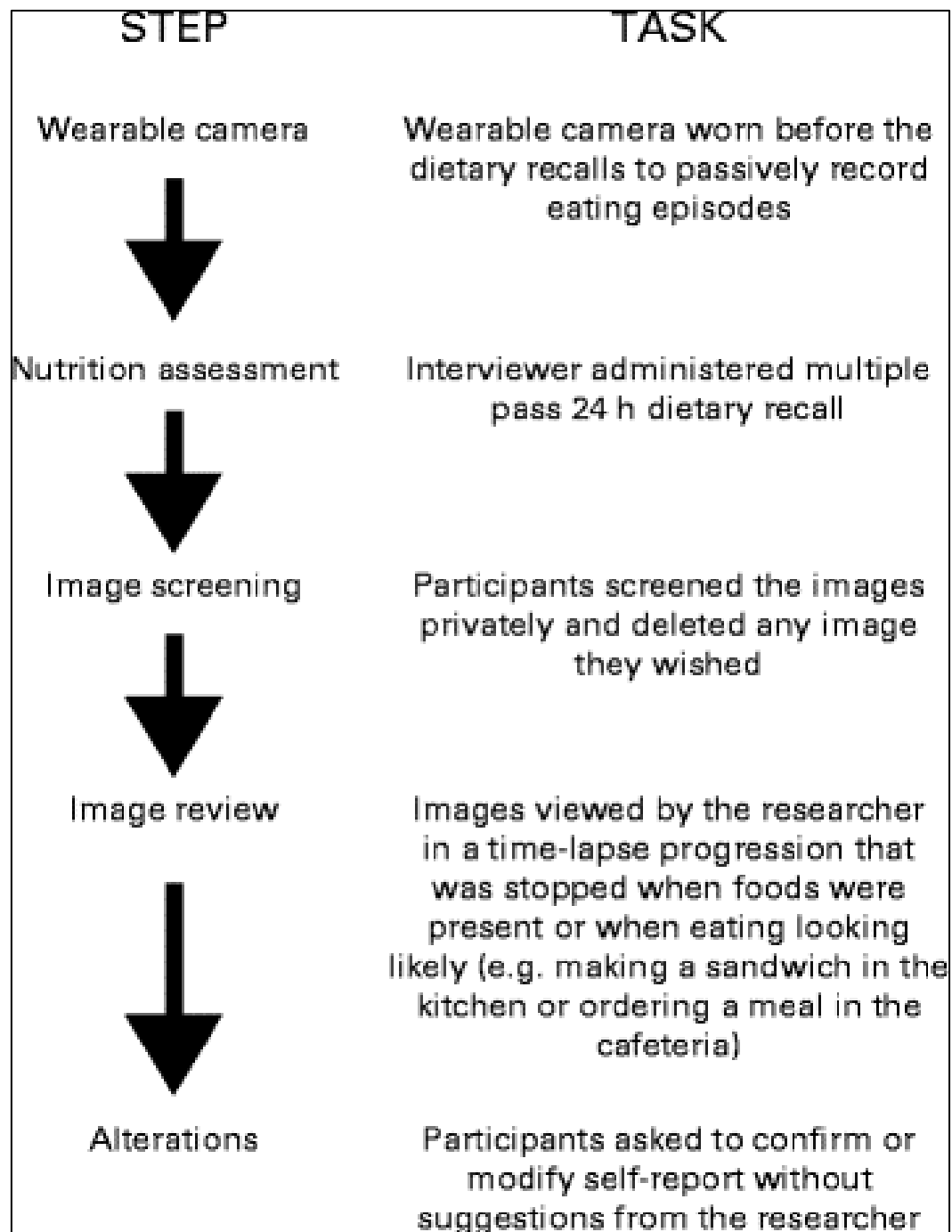




Source: Orcholski et al. (2015)



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Wearable cameras reduced the magnitude of underreporting by 8% in men and 6% for women

Source: Gemming et al. (2014)

Q: Could wearable cameras affect intake?

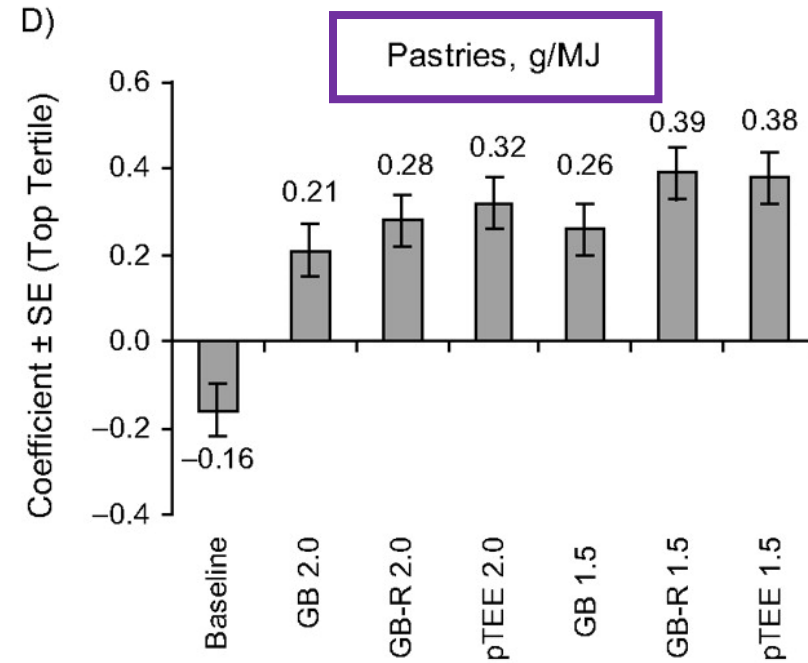
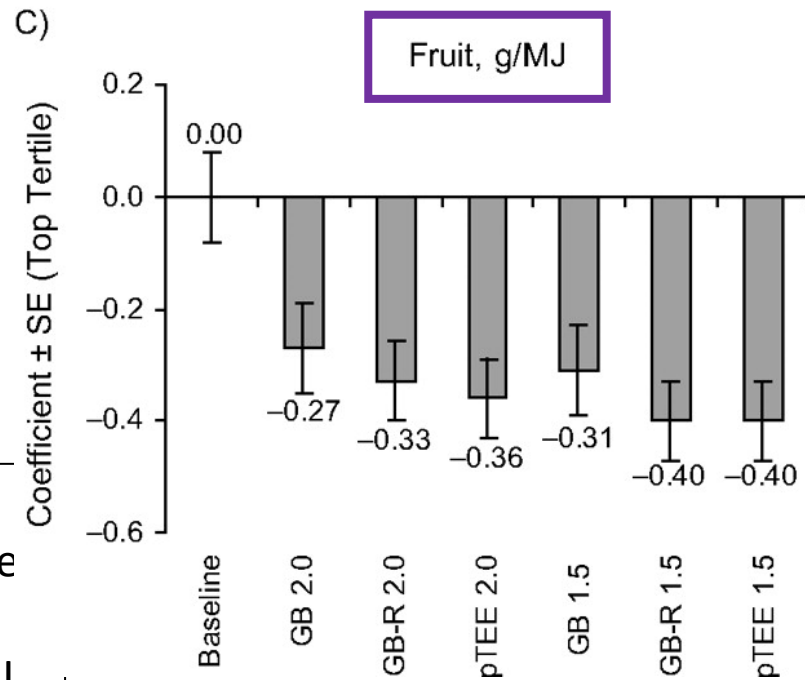
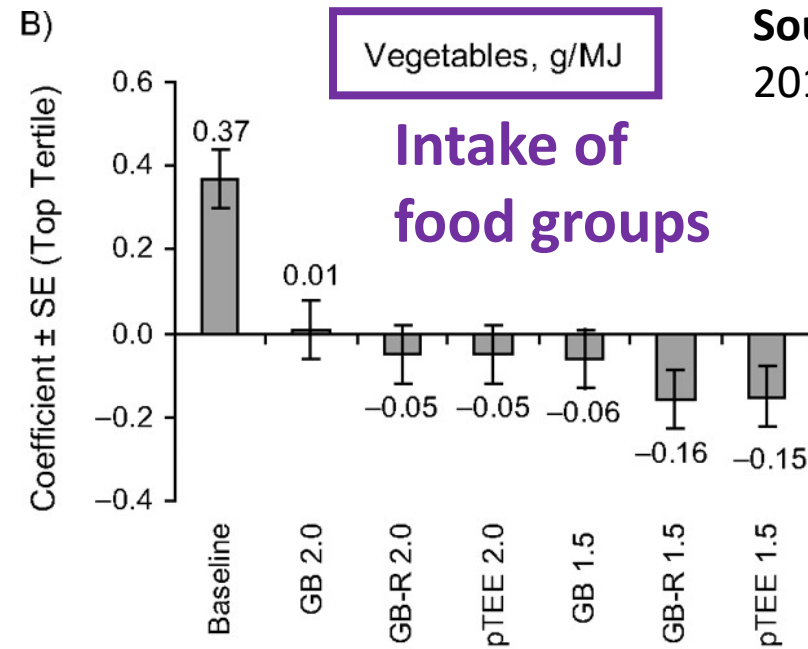
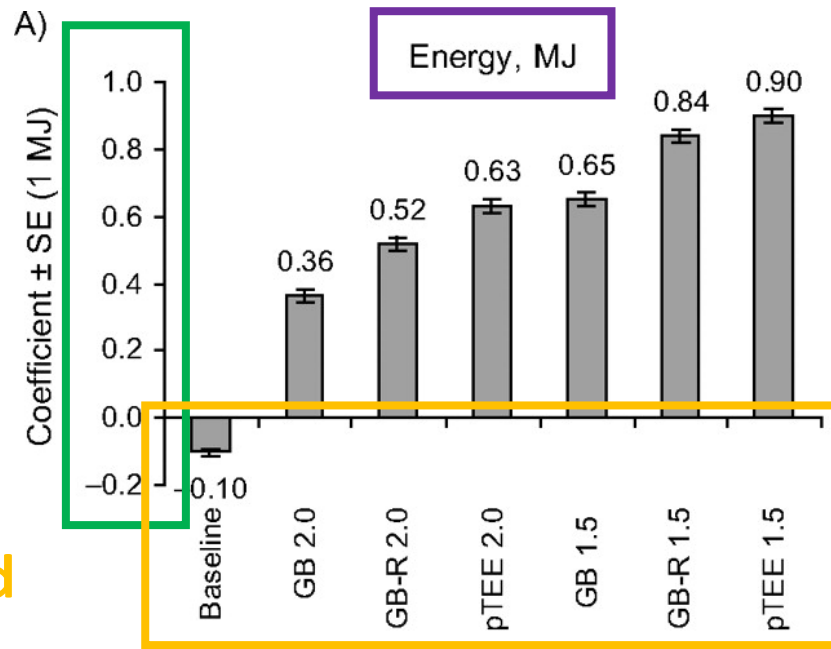


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Coefficient of intake on BMI

Adjusting the estimated coefficients based on different methods for verifying intake.

Adjustment generally **increases** the magnitude of estimated relations between intake and BMI



Source: Mendez et al. 2011 (Fig. 1, women)

Underreporting with Intake24

- Sample of 98 UK adults aged 40-65 years
- Compared with measured total energy expenditure (TEE) using doubly-labeled water, Intake24 underestimated energy intake by 22-25%
- Energy intake from Intake24 is positively correlated with TEE from doubly labeled water
- Energy intake from Intake24 is similarly underestimated as intake estimated from interviewer-led 24-hr recalls and food diaries

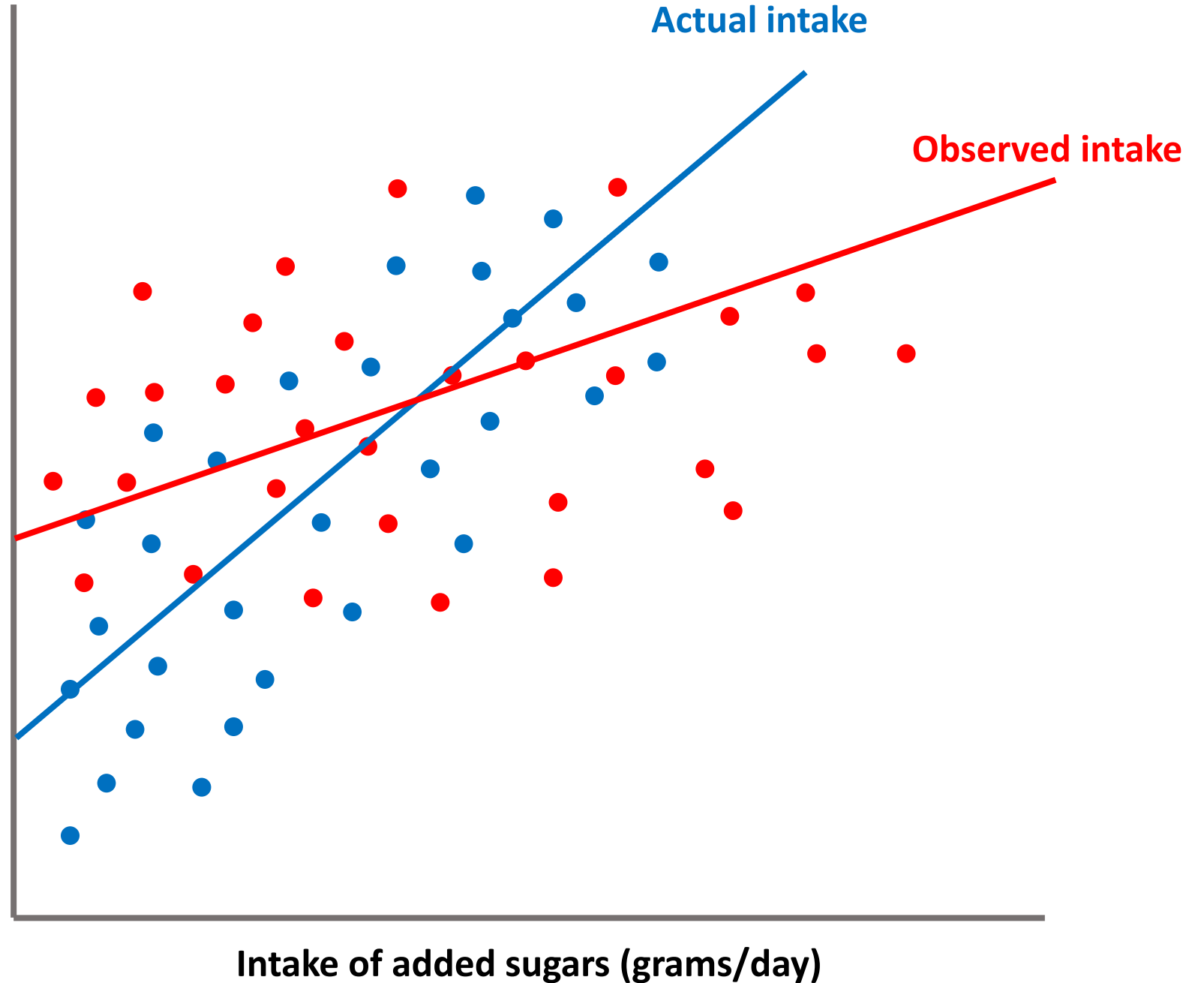
Q: What does this finding tell us?

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Hgb A1c (%)

Q: What if intake of added sugar tended to be underreported when people consumed larger amounts, and overreported when people tended to eat smaller amounts?



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Types of missing data

- **Missing completely at random (MCAR)**
 - True random missingness
- **Missing at random (MAR)**
 - Random missingness can be fully explained with observable factors (but not the missing variable itself)
 - Can't prove this, but one can do statistical analysis using these data
- **Missing not at random (MNAR)**
 - Differences between missing and non-missing remains after taking observable differences into account
 - Can prove that data is MNAR if missingness is not fully explained by observable factors

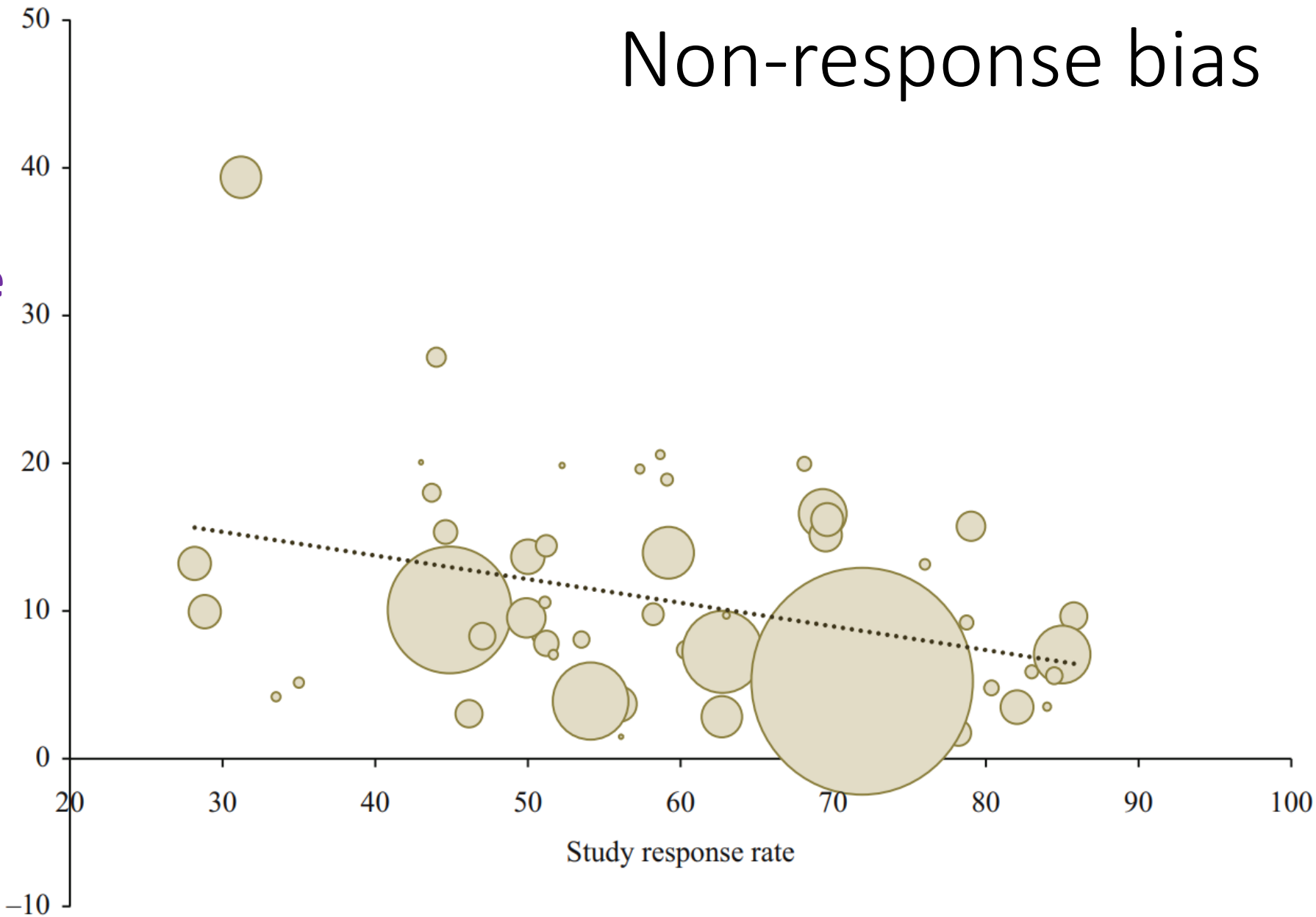


Non-response bias

Increasing study response rates reduces nonresponse bias.

Size of bubble = sample size in original study

Relative bias statistic for a given study



Ways to reduce non-response bias

- Keep surveys **short**
- Keep surveys **simple**
- Provide an **incentive?**
- Gentle **reminders**
- **Follow-up**
- **Weighting**
 - Intake24 data was "weighted" on age and gender to ensure that, across those dimensions, the sample was representative of the population.
 - Data were also weighed by household size, because people in larger households are less likely to be selected.

Q: What are some barriers or tradeoffs to using these strategies?



On weighting

- Weighting means **assigning values for the relative contribution** of different factors in the calculation of a population-level statistic
 - A “statistic” is something like “average daily fluid milk intake.”
- When survey results are **weighted**, each respondent's data might count for relatively more or less depending on some observable characteristics like age, sex, gender, household size, or income.
- Weighting **can** restore the representativeness of a sample, but can only restore representativeness on observable factors
 - Observable factors may be affected by **measurement error**
- Can't weight on what can't be observed
 - By definition



Qs: Imagine you are sitting down to analyze new dietary intake data from a study

- If non-respondents had actually responded, can we assume that their estimates would be similar to respondents? Why or why not?
- What factors might influence the likelihood of responding? Are these factors also related to dietary intake, or health status?

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Ways to improve data quality

Before and during data collection

- Training enumerators
- Using 'hard' measures vs. soft
- Validate survey instruments
- Interview mode
- Enumerator gender
- Question wording
- Question order
- Reduce survey fatigue
- Deal with heaping
 - People defaulting to nice round #s
- Understand incentives for respondents and enumerators

During data cleaning and analysis

- Cross-check
- Spot check
- Check for biological plausibility, examine outliers
- Estimate reliability of instruments
- Perform sensitivity analysis
- Account for sampling design
- Estimate design-based sampling errors
- Imputation
- Weighting
- Assess heaping

FSS already engages in many data quality checks

- Nutritionist examines data for mistakes
- Manual coding as needed
- Amounts checked for “reasonableness”
- Histograms to check distributions and extreme intakes
- Examine recalls completed in 2 min or less
- Collected respondent feedback – quant and qual, in Pilot
- Collected interviewer feedback – quant and qual, in Pilot



Qs: Given what we all know about diet assessment...

- Why do we still want to assess dietary intake, when a lot can go wrong?
- Is it worth it? Why or why not?



References

- Brick, J. Michael, and Roger Tourangeau. "Responsive survey designs for reducing nonresponse bias." *Journal of Official Statistics* 33, no. 3 (2017): 735-752.
- Foster, E., Lee, C., Imamura, F., Hollidge, S.E., Westgate, K.L., Venables, M.C., Poliakov, I., Rowland, M.K., Osadchiy, T., Bradley, J.C. and Simpson, E.L., 2019. Validity and reliability of an online self-report 24-h dietary recall method (Intake24): a doubly labelled water study and repeated-measures analysis. *Journal of nutritional science*, 8.
- Gemming, L., Rush, E., Maddison, R., Doherty, A., Gant, N., Utter, J. and Mhurchu, C.N., 2015. Wearable cameras can reduce dietary under-reporting: doubly labelled water validation of a camera-assisted 24 h recall. *British Journal of Nutrition*, 113(2), pp.284-291.
- Groves, R.M. and Lyberg, L., 2010. Total survey error: Past, present, and future. *Public opinion quarterly*, 74(5), pp.849-879.
- Mendez, M.A., Popkin, B.M., Buckland, G., Schroder, H., Amiano, P., Barricarte, A., Huerta, J.M., Quirós, J.R., Sánchez, M.J. and González, C.A., 2011. Alternative methods of accounting for underreporting and overreporting when measuring dietary intake-obesity relations. *American journal of epidemiology*, 173(4), pp.448-458.
- Orcholski, L., Luke, A., Plange-Rhule, J., Bovet, P., Forrester, T.E., Lambert, E.V., Dugas, L.R., Kettmann, E., Durazo-Arvizu, R.A., Cooper, R.S. and Schoeller, D.A., 2015. Under-reporting of dietary energy intake in five populations of the African diaspora. *British Journal of Nutrition*, 113(3), pp.464-472.